AR/VR MCQs with Correct Answers

Q1. Mixed Reality (MR) differs from AR because in MR virtual objects:

(a) Fully replace the real world (b) Can interact with and respond to the real environment (c) Work only on phones (d) Need no camera

Answer: (b) Can interact with and respond to the real environment

Q2. Images projected onto a building during a light show are an example of:

(a) Marker-based AR (b) Marker-less AR (c) Projection Mapping (d) Virtual Reality

Answer: (c) Projection Mapping

Q3. Placing a virtual sofa on your floor without any printed image is an example of:

(a) Marker-based AR (b) Marker-less AR (c) Projection Mapping (d) VR

Answer: (b) Marker-less AR

Q4. Which technology usually has the lower hardware barrier for users?

(a) Mobile AR (b) PC-tethered VR (c) Both are equal (d) Neither needs hardware

Answer: (a) Mobile AR

Q5. A VR headset that lets you walk, crouch and lean inside a game supports:

(a) 2DOF (b) 3DOF (c) 6DOF (d) 0DOF

Answer: (c) 6DOF

Q6. Which of these is NOT a rotational movement?

(a) Pitch (b) Yaw (c) Roll (d) Moving forward

Answer: (d) Moving forward

Q7. Lens Studio is available to creators as:

(a) A paid subscription (b) A free tool (c) A tool only for Snap employees (d) A browser plugin only

Answer: (b) A free tool

Q8. Which helps a beginner quickly make a popular selfie effect?

(a) Face Templates (b) Terminal (c) Compiler (d) Debug console

Answer: (a) Face Templates

Q9. Which is NOT a place where a Lens Studio Lens can run?

(a) Snapchat (b) Spectacles (c) Apps using Camera Kit (d) A gaming console

Answer: (d) A gaming console

Q10. Camera Kit lets developers:

(a) Bring Snap AR Lenses into their own mobile and web apps (b) Edit videos (c) Design 2D logos (d) Build VR games

Answer: (a) Bring Snap AR Lenses into their own mobile and web apps

Q11. Where do you browse ready-made 3D models, scripts and packages from Snap and the community?

(a) Inspector (b) Asset Library (c) Scene Hierarchy (d) Preview panel

Answer: (b) Asset Library

Q12. A Lens that reacts when the user opens their mouth relies on:

(a) World tracking (b) Face tracking (c) Marker tracking (d) Hand tracking

Answer: (b) Face tracking

Q13. A Lens that draws a glowing outline around a dancer's full body uses:

(a) Face tracking (b) Hand tracking (c) Body tracking (d) Marker tracking

Answer: (c) Body tracking

Q14. A Lens that detects a thumbs-up gesture uses:

(a) Body tracking (b) Hand tracking (c) World tracking (d) Face tracking

Answer: (b) Hand tracking

Q15. Which GenAI tool creates a 3D model from a text description?

(a) 3D Asset Generation (b) Marker tracking (c) Orthographic camera (d) Behavior script

Answer: (a) 3D Asset Generation

Q16. CharacterAnimation in the GenAI Suite is used to:

(a) Add motion to a 3D character (b) Change the camera type (c) Reduce Lens size (d) Record a preview video

Answer: (a) Add motion to a 3D character

Q17. Marker tracking needs:

(a) A reference image (b) GPS (c) A microphone (d) A VR headset

Answer: (a) A reference image

Q18. Bitmoji support in Lens Studio lets a Lens:

(a) Show the user's own avatar (b) Track hands (c) Detect markers (d) Edit audio

Answer: (a) Show the user's own avatar

Q19. Which component plays background music in a Lens?

(a) Audio component (b) Text component (c) Image component (d) Light

Answer: (a) Audio component

Q20. Which camera gives a 3D car model realistic depth?

(a) Perspective (b) Orthographic (c) Both equally (d) Neither

Answer: (a) Perspective

Q21. The Tween script is mainly used to:

(a) Animate position, scale, rotation or colour over time (b) Track the face (c) Publish a Lens (d) Import 3D models

Answer: (a) Animate position, scale, rotation or colour over time

Fill in the Blanks with Answers

1. _____ AR uses surface detection and device sensors instead of a printed image.

Answer: Marker-less

2. In VR, the user loses _____ awareness of their real surroundings.

Answer: Visual

3. Nodding your head “yes” is a rotation called _____.

Answer: Pitch

4. Shaking your head “no” is a rotation called _____.

Answer: Yaw

5. Tilting your head towards your shoulder is a rotation called _____.

Answer: Roll

6. 6DOF adds _____ movement to the three rotational movements of 3DOF.

Answer: Translational

7. Snap’s AR glasses, for which Lenses can also be built, are called _____.

Answer: Spectacles

8. Ready-made starter projects in Lens Studio are called _____ Projects.

Answer: Template

9. The small picture that represents a Lens to users is called the Lens _____.

Answer: Icon

10. Hand tracking follows points on the hand called hand _____.

Answer: Joints

11. Face tracking places virtual glasses correctly by following facial _____ such as the eyes and nose.

Answer: Landmarks

12. World _____ builds a 3D model of the surroundings so real objects can hide virtual ones.

Answer: Mesh

13. 3D Body _____ is a GenAI effect that changes the shape of the user’s body.

Answer: Morph

14. Immersive Machine Learning effects are part of Lens Studio’s _____ Suite.

Answer: GenAI

15. The printed image that triggers content in marker tracking is called a _____.

Answer: Marker

16. Lens Studio’s _____ features let a Lens show the user’s own avatar instead of a generic character.

Answer: Bitmoji

17. To show a 2D logo on the screen, you add a Screen _____ to the scene.

Answer: Image

18. In the _____ Camera, two railway tracks appear to meet in the distance.

Answer: Perspective

19. A Behavior script can run a response when the user _____ the screen.

Answer: Taps

20. A Tween can be set to _____ so the animation repeats continuously.

Answer: Loop

AR/VR Short Questions (2 marks each)

1. In your own words, explain what "overlaying digital information on the real world" means. Give an everyday example.

Answer: It means adding digital objects, images, or information to the real-world view using AR technology.

Example: Using a mobile AR app to see a virtual sofa in your living room.

2. Give one example each of AR, VR and MR used in education.

Answer:

- AR: Viewing a 3D model of the human heart on a textbook.
- VR: Exploring a virtual science laboratory.
- MR: Interacting with a virtual 3D human skeleton in a real classroom.

3. Name two industries that use XR today and how they use it.

Answer:

1. Healthcare: Used for medical training and surgical practice.
2. Education: Used for interactive learning and virtual classrooms.

4. Why is marker-less AR more flexible than marker-based AR?

Answer: Marker-less AR uses sensors and surface detection instead of a printed marker. It allows users to place virtual objects on suitable real-world surfaces without needing a specific image.

5. Why does VR usually need more computing power than mobile AR?

Answer: VR must render a complete virtual environment, often for both eyes, at a high frame rate. This requires more processing power than many mobile AR applications.

6. Name two devices used for AR and two used for VR.

Answer:

- AR devices: Smartphone, AR glasses.
- VR devices: Meta Quest headset, HTC Vive headset.

7. Draw and label the six degrees of freedom of a head-mounted device.

Answer: The six degrees of freedom (6DOF) include three rotational movements and three translational movements.

6 Degrees of Freedom (6DOF)

Rotational movements (3)

- Pitch: Head moves up and down.
- Yaw: Head turns left and right.
- Roll: Head tilts towards either shoulder.

Translational movements (3)

- X-axis: Moves left and right.
- Y-axis: Moves up and down.
- Z-axis: Moves forward and backward.

8. Give one example of an experience that needs only 3DOF and one that needs 6DOF.

Answer:

- 3DOF: Looking around a 360° virtual video while standing in one place.
- 6DOF: Walking, crouching, and leaning inside a VR game.

9. What is spatial awareness? Why does it matter for an AR navigation app?

Answer: Spatial awareness is the ability of a device or system to understand the position, distance, and layout of objects in the real environment.

It helps an AR navigation app place arrows and directions correctly on roads and paths.

10. What are Sample Projects? How do they help a beginner?

Answer: Sample Projects are ready-made example projects that demonstrate how to create AR experiences.

They help beginners understand project structure, learn features, explore working examples, and create their own Lenses more easily.

11. Give one example of a business that could use Camera Kit, and why.

Answer: An online shopping business can use Camera Kit to add AR try-on features to its app, allowing customers to preview products before buying.

12. What is the difference between a selfie (front camera) Lens and a world (rear camera) Lens?

Answer:

- Selfie Lens: Uses the front camera to apply effects to the user's face.
- World Lens: Uses the rear camera to place virtual objects in the real environment.

13. Why should a Lens be kept small in size? Give two ways to reduce its size.

Answer: A small Lens downloads faster and provides a smoother user experience.

Two ways to reduce its size:

1. Compress images and textures.
2. Remove unused assets and reduce 3D model complexity.

14. List any four details you prepare before submitting a Lens.

Answer:

1. Lens name and description.
2. Lens icon.
3. Preview image or video.
4. Relevant category and other required publishing details.

15. Mention two benefits a creator gets from publishing Lenses for Snapchatters.

Answer:

1. Reaches a wider audience through Snapchat.

2. Builds creativity, visibility, and recognition as an AR creator.

16. How does the Asset Library save time compared with building every 3D model yourself? Give an example.

Answer: The Asset Library provides ready-made 3D models, scripts, and other resources, so creators do not need to build everything from scratch.

Example: Using a ready-made 3D flower model instead of creating one manually.

17. Why should a creator check the size of an asset before adding it to a Lens?

Answer: Large assets increase the Lens file size and may affect loading speed and performance. Checking asset size helps keep the Lens lightweight and smooth.

18. What is occlusion in AR? How does World Mesh help with it?

Answer: Occlusion occurs when a real object hides a virtual object from view.

World Mesh represents the surrounding environment in 3D, helping virtual objects appear behind real surfaces when the appropriate occlusion setup is used.

19. What is surface (ground plane) detection in world tracking?

Answer: Surface detection identifies suitable real-world surfaces, such as floors, tables, and ground. It allows users to place virtual objects accurately on those surfaces.

20. Give one Lens idea that combines two tracking types, for example face and hand.

Answer: A magic makeup Lens can combine face tracking and hand tracking. Face tracking applies makeup effects, while hand tracking lets the user control the effects using hand gestures.

21. List three uses of body tracking in fitness or fashion.

Answer:

1. Tracking body movements during exercise.
2. Checking exercise posture and body position.
3. Virtually previewing clothes on a user's body.

22. Why is the GenAI Suite helpful for creators who are not 3D artists?

Answer: The GenAI Suite helps creators generate 3D assets and animations using AI tools. It reduces the need for advanced 3D modelling and animation skills.

23. Why can marker tracking fail with a plain white image or in low light?

Answer: Marker tracking needs clear visual features to recognise a reference image. A plain white image has very few features, while low light makes the image difficult for the camera to detect.

24. What happens to the AR content when the marker moves out of the camera view?

Answer: The system may lose track of the marker, causing the attached AR content to disappear, freeze, or stop following the marker.

25. Give two Lens ideas that use the user's 3D Bitmoji avatar.

Answer:

1. A birthday Lens showing the user's Bitmoji dancing beside a virtual cake.
2. A celebration Lens showing the Bitmoji performing animated actions.

26. Why should audio clips in a Lens be short and compressed?

Answer: Short, compressed audio clips reduce Lens file size, load faster, and use less memory, helping the Lens run smoothly.

27. Give one example where a Lens should play audio only after the user does something, instead of at the start.

Answer: In a birthday Lens, birthday music should play only when the user taps the virtual cake, rather than playing automatically when the Lens opens.

28. Why does on-screen Screen Text stay the same size when the user moves the phone away?

Answer: Screen Text is attached to the screen rather than positioned as a real-world 3D object. Therefore, its displayed size remains approximately constant as the phone moves.

29. List any four triggers available in the Behavior script.

Answer:

1. Screen tap.
2. Touch or interaction event.
3. Object appearing or becoming enabled.
4. A timer or delay event.

Note: Exact trigger names depend on the Behavior system and Lens Studio version.

30. Name two kinds of Tween you can apply to an object.

Answer:

1. Position Tween: Moves an object smoothly from one position to another.
2. Scale Tween: Changes an object's size smoothly over time.

31. Why is it useful to set up interactions without writing code?

Answer: It allows beginners to create interactions using visual settings and predefined actions. It saves development time and reduces coding errors.

Long Questions:

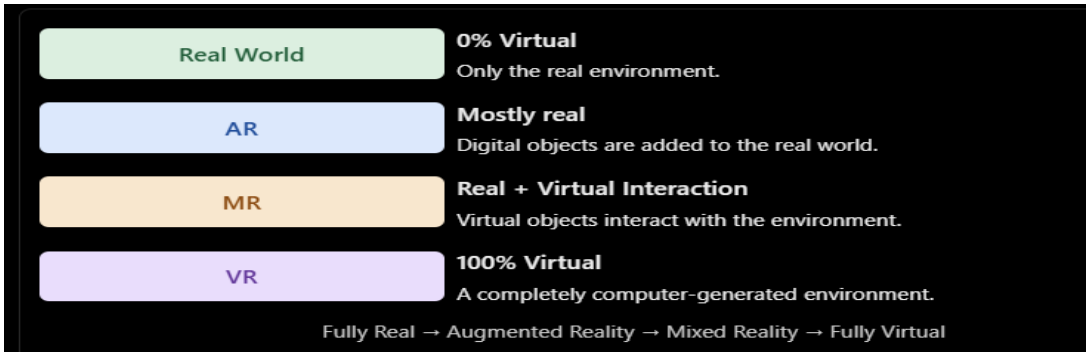
Q1. Arrange the real world, AR, MR and VR on a scale from fully real to fully virtual. Explain each point on the scale with an example.

Definition

The Reality–Virtuality Continuum is a scale that ranges from the completely real environment to a completely virtual environment.

Reality–Virtuality Scale

Real World



Explanation

1. Real World: The user experiences the physical environment without digital additions. Example: Reading a printed textbook.

2. Augmented Reality (AR): Digital objects or information are overlaid on the real world. Example: Viewing a virtual sofa through a mobile camera.

3. Mixed Reality (MR): Virtual objects are placed in the real environment and can interact with real surfaces. Example: A virtual ball bouncing off a real table.

4. Virtual Reality (VR): The user enters a fully computer-generated environment, usually using a VR headset. Example: Exploring a virtual museum.

Key point: Moving from the real world towards VR increases the amount of computer-generated content.

Q2. Compare marker-based and marker-less AR: how each decides where to place content, their advantages, limitations and one example each.

Definition

Marker-based AR uses a predefined image or visual marker to detect where digital content should appear. Marker-less AR uses technologies such as surface detection, device sensors, and environmental tracking to place digital content without a printed marker.

Comparison Table

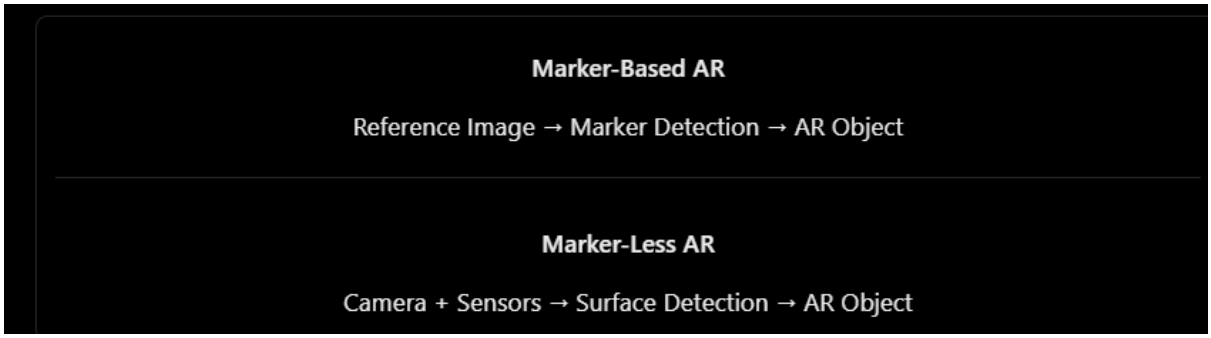
Point	Marker-Based AR	Marker-Less AR
Placement	Detects a predefined image or marker.	Detects surfaces and the surrounding environment.
Requirements	Needs a suitable reference marker.	Does not need a printed marker.
Advantages	Simple, accurate placement when the marker is detected clearly.	Flexible placement on suitable real-world surfaces.
Limitations	Can fail if the marker is hidden, unclear, or poorly lit.	May struggle with poor lighting or surfaces with few visual features.
Example	Scanning a QR-like image in a book to display a 3D model.	Placing a virtual sofa on the living-room floor using a phone.

Working

1. Marker-Based AR: The camera captures the marker, the system recognises it, and digital content is placed relative to it.

2. Marker-Less AR: The camera and supported sensors analyse the environment, detect suitable surfaces, and place digital objects in the selected location.

Simple Diagram



Remember for the exam: Marker-based AR depends on a reference image, while marker-less AR places content using environmental tracking and surface detection.

Q3. Explain how projection mapping works and give two real-world uses. How is it different from phone-based AR?

Definition

Projection mapping is a technology that projects images, videos, or animations onto physical surfaces such as buildings, walls, and stages.

How Projection Mapping Works

1. Select a surface: Choose a building, wall, stage, or other physical object.
2. Map the surface: Measure or scan its shape and dimensions.
3. Prepare visuals: Create images, videos, or animations that fit the surface.
4. Align the projector: Adjust the projected visuals to match the surface accurately.
5. Display content: Project the visuals to create an interactive or realistic visual experience.

Two Real-World Uses

1. Entertainment: Project colourful animations onto buildings during festivals and light shows.
2. Events and Advertising: Project videos and visual effects onto stage backgrounds or product displays to attract audiences.

Projection Mapping vs Phone-Based AR

Point	Projection Mapping	Phone-Based AR
Technology	Uses a projector.	Uses a smartphone camera and screen.
Display	Projects visuals onto physical surfaces.	Displays digital objects on the phone screen over the camera view.
Example	Animated light show on a building.	Viewing a virtual sofa in a room.

Key point: Projection mapping puts light and images directly onto physical surfaces, whereas phone-based AR displays digital content through a device screen.

Q4. A training institute must choose between AR and VR for (a) a fire-safety drill and (b) machine repair on a factory floor. Recommend one for each case and justify.

Definition

AR (Augmented Reality) adds digital information to the real environment, while VR (Virtual Reality) creates a computer-generated environment for an immersive experience.

(a) Fire-Safety Drill — VR

Recommendation: Virtual Reality (VR)

1. Safe practice: Trainees can practise emergency situations without facing real fire.
2. Realistic simulation: VR can simulate smoke, fire, and evacuation scenarios.
3. Risk-free learning: Trainees can learn how to respond to dangerous situations.
4. Repeat training: The same drill can be repeated to improve emergency response.

Example: Trainees wear VR headsets and practise escaping from a virtual burning building.

(b) Machine Repair on a Factory Floor — AR

Recommendation: Augmented Reality (AR)

1. Real machine visibility: Workers can see the actual machine while repairing it.
2. Step-by-step guidance: AR can display repair instructions over the machine.
3. Part identification: Digital labels can identify components, bolts, and tools.
4. Less confusion: Visual instructions help workers follow the correct repair procedure.

Example: A technician uses an AR-enabled phone or headset to view instructions for replacing a machine part.

Quick Comparison

Requirement	Best Choice	Reason
Fire-safety drill	VR	Safe simulation of dangerous emergencies.
Factory machine repair	AR	Shows instructions on the real machine.

Q5. Explain why AR suits retail and navigation, while VR suits gaming and training. Give two examples of each.

Definition

Augmented Reality (AR) adds digital information to the real world, while Virtual Reality (VR) creates a completely computer-generated environment.

1. Why AR Suits Retail and Navigation

1. Real-world view: AR allows users to see their actual surroundings.
2. Product preview: Customers can view virtual products in their homes before buying.
3. Live directions: AR can display arrows and directions over the camera view.
4. Easy access: Many AR applications work on smartphones.

Examples:

- Retail: Previewing a virtual sofa in a living room before purchasing it.
- Navigation: Following AR arrows to find a destination.

2. Why VR Suits Gaming and Training

1. Immersion: VR makes users feel present inside a virtual environment.
2. Realistic experience: It creates simulated environments for games and practice.

3. Safe training: Users can practise dangerous tasks without real-world risks.
4. Interaction: Users can explore and interact with virtual objects.

Examples:

- Gaming: Playing a virtual adventure game using a VR headset.
- Training: Practising a flight simulation without flying a real aircraft.

Quick Comparison

AR	VR
Adds digital content to the real world.	Creates a virtual environment.
Best suited for retail and navigation.	Best suited for immersive gaming and training.
Commonly uses smartphones.	Commonly uses VR headsets and controllers.

Q6. A treasure-hunt game could be built in VR or in AR. Compare both versions in terms of immersion, hardware, safety and reach.

Definition

An AR treasure hunt places virtual clues and treasures in the real environment. A VR treasure hunt places the player inside a completely virtual world to search for clues and treasures.

Comparison Table

Point	AR Treasure Hunt	VR Treasure Hunt
Immersion	Combines virtual clues with the real world.	Provides a more immersive, fully virtual experience.
Hardware	Usually requires a smartphone or AR glasses.	Usually requires a VR headset and possibly controllers.
Safety	Players must watch their surroundings while moving.	Players may collide with nearby objects if the play area is unsafe.
Reach	Can be played in parks, campuses, and public places using compatible devices.	Can be played indoors by people with compatible VR equipment.
Experience	Players find virtual treasures at real-world locations.	Players explore virtual caves, castles, or fantasy worlds.

Examples

1. AR Version: Students walk around their college campus, follow digital clues, and discover virtual treasure chests through their phones.
2. VR Version: Players explore a virtual castle, solve puzzles, and collect hidden treasure using a VR headset.

Recommendation

- Choose AR for a campus or outdoor treasure hunt because it encourages real-world exploration and can reach smartphone users.
- Choose VR for a fantasy adventure with immersive environments, puzzles, and virtual challenges.

Q7. Explain the kinds of experiences Lens Studio can build: selfie Face Lenses, shoppable try-on, world Lenses and Spectacles experiences. Give one example each.

Definition

Lens Studio is Snap's free AR development tool used to create interactive augmented reality experiences for Snapchat and other supported platforms.

Types of Experiences

1. Selfie Face Lenses: These use face tracking to add digital effects, makeup, masks, or accessories to a user's face. Example: A Lens that adds virtual sunglasses and a moustache to the user's face.

2. Shoppable Try-On: These allow users to preview products virtually before purchasing them. Example: Trying virtual spectacles or sunglasses through a smartphone camera.

3. World Lenses: These place digital objects, animations, or effects into the user's real surroundings using the camera. Example: Placing a virtual 3D dinosaur on the floor of a room.

4. Spectacles Experiences: These are AR experiences designed for supported Snap Spectacles devices, allowing digital content to appear in the user's view of the real world. Example: Viewing virtual directions or interactive 3D objects through compatible Spectacles.

Quick Comparison

Experience	Main Purpose	Example
Face Lens	Face effects	Virtual sunglasses
Shoppable try-on	Product preview	Trying spectacles
World Lens	Add objects to surroundings	3D dinosaur
Spectacles experience	Wearable AR	Virtual directions

Key point: Lens Studio supports different AR experiences for entertainment, shopping, real-world interaction, and compatible wearable devices.

Q8. How do Lens Studio's "zero setup" and direct Snapchat integration help a new AR creator? Explain with an example.

Definition

Zero setup means creators can get started with minimal configuration by using ready-made templates and resources. Direct Snapchat integration allows creators to test and publish supported Lenses for Snapchat users through Snap's workflow.

Main Points

1. Easy to start: Beginners can use Lens Studio without building an AR project completely from scratch.

2. Ready-made templates: Templates provide pre-built effects, tracking features, and project structures.

3. Quick testing: Creators can preview and test their Lenses during development.

4. Direct integration: Supported workflows connect Lens creation with Snapchat, making it easier to share published Lenses.

5. Less development time: Beginners can focus on designing creative effects instead of spending too much time on initial setup.

Example

A beginner wants to create a birthday Face Lens.

1. The creator opens Lens Studio and selects a suitable Face Lens template.

2. They add birthday text, balloons, and a face effect.
3. They preview and test the Lens.
4. They submit it through the publishing process.
5. After approval, Snapchat users can access the published Lens.

Q9. Explain how a creator can preview and test a Lens on a real phone before publishing. Why is testing on a device important?

Definition: Previewing and testing a Lens means checking its appearance, tracking, and performance on a real smartphone before publishing it.

Steps to preview and test a Lens:

1. Create the Lens: Design the AR effect in Lens Studio and save the project.
2. Connect to Snapchat: Use the available device-preview or Snapchat testing workflow to open the Lens on a compatible phone.
3. Check tracking: Move the phone and face to check whether the Lens follows correctly.
4. Test interactions: Try taps, gestures, animations, and audio if included.
5. Check performance: Observe loading speed, smoothness, and battery or device performance.
6. Fix errors: Correct any problems and test the Lens again before submitting it.

Why is device testing important?

- It checks how the Lens works in real-world conditions.
- It helps find tracking and interaction errors.
- It ensures the Lens runs smoothly on the target device.
- It checks that text, images, and 3D objects appear correctly.
- It improves the user experience before publishing.

Example: A creator makes a birthday Face Lens with balloons and music. Testing on a phone checks whether the balloons follow the face correctly and the music plays as expected.

Q10. Explain good practices to follow before submitting a Lens: size, icon, preview, testing and original content.

Definition: Good submission practices help ensure that a Lens is optimized, clearly presented, properly tested, and suitable for publishing.

Important practices:

1. Optimize size: Compress images and audio, remove unused assets, and reduce unnecessary 3D model complexity. This helps the Lens load faster.
2. Create a clear icon: Use a simple, attractive icon that represents the Lens and is easy to recognize.
3. Prepare a preview: Provide a suitable preview image or video showing the main AR effect clearly.
4. Test the Lens: Check tracking, animations, interactions, audio, and performance on a real device. Fix errors before submission.
5. Use original content: Create your own designs or use assets you have permission to use. Avoid copying copyrighted images, music, logos, or models without authorization.
6. Check publishing details: Enter the correct Lens name, description, and other required information before submitting.

Example: Before submitting a festival Lens, a creator compresses its graphics, creates a matching icon, records a preview, tests the effect on a phone, and checks that all music and images are properly licensed.

Q11. Explain face tracking in Lens Studio: what it detects and how effects such as makeup, masks and face distortion use it. Give an example.

Definition: Face tracking is an AR technique that detects a person's face and follows its position and movements in real time.

Working and uses:

1. Face Detection: Identifies the face in the camera view.
2. Facial Landmarks: Detects important facial points such as the eyes, nose, lips, eyebrows and jawline.
3. Makeup Effects: Adds virtual lipstick, eyeliner, blush and other makeup to the correct facial areas.
4. Mask Effects: Places a virtual 2D or 3D mask over the face and adjusts it when the user moves.
5. Face Distortion: Changes facial features, such as enlarging the eyes or nose, to create funny effects.
6. Real-Time Tracking: Keeps the effects aligned with the face while the user turns their head, smiles or blinks.

Example: A Snapchat Lens adds virtual lipstick and a cat mask to the user's face. When the user turns their head, the effects move along with the face.

Q12. Explain body tracking with two use cases: a virtual fashion try-on and a fitness coach Lens.

Definition: Body tracking is an AR technique that detects and follows a person's body position and movements using a camera.

Use case 1: Virtual Fashion Try-On

1. Body Detection: Detects the user's body and its position.
2. Virtual Clothing: Places digital clothes, such as shirts, jackets or dresses, over the body.
3. Movement Tracking: Adjusts the virtual clothing as the user moves.
4. Benefit: Allows users to preview outfits without physically wearing them.

Example: A user previews a virtual jacket through an AR Lens before buying it.

Use case 2: Fitness Coach Lens

5. Movement Tracking: Tracks body movements during exercises such as squats and lunges.
6. Visual Guidance: Displays instructions or feedback to help users improve their exercise posture.

Example: A fitness Lens tracks a user's squats and displays guidance when their body position needs adjustment.

Key difference: Fashion try-on displays virtual clothing on the body, while a fitness coach tracks movements to support exercise guidance.

Q13. In a Lens game, virtual balloons float around the room and the user pops them by pointing a finger. Which tracking types are needed and why?

Definition: Tracking in AR identifies and follows real-world objects, body parts, or movements so virtual objects can interact with the user.

Required tracking types:

1. World Tracking: Tracks the phone's movement and position in the room so virtual balloons appear in the environment.

2. Surface Detection: Detects floors, tables, or other suitable surfaces to place or position virtual objects correctly.
3. Hand Tracking: Detects the user's hand and finger positions to identify pointing gestures.
4. Gesture Recognition: Detects the pointing or tapping gesture to trigger the balloon-popping action.
5. Collision or Interaction Detection: Checks whether the pointing action reaches a balloon and triggers its popping animation.
6. Visual Feedback: Plays a popping animation or sound when the balloon is successfully hit.

Example: The user points their index finger at a floating balloon. The Lens detects the gesture, registers the hit, and removes the balloon with a popping animation.

Flow diagram:

World Tracking → Balloon Placement → Hand Tracking → Pointing Detection → Balloon Pops

Q14. Explain how a creator could use 3D Asset Generation to build a Lens quickly. What are its limitations?

Definition: 3D Asset Generation is an AI-assisted feature that helps creators generate 3D objects from prompts or other supported inputs for use in AR experiences.

How it helps create a Lens:

1. Enter a Prompt: Describe the required object, such as “a cute 3D birthday cake.”
2. Generate the Model: Use the available AI tool to create a 3D asset.
3. Import into Lens Studio: Add the generated model to the Lens project.
4. Customize the Asset: Adjust its size, position, materials, lighting, and animation as needed.
5. Build the Lens: Combine the model with tracking, interactions, and visual effects.
6. Test the Result: Preview the Lens on a device and optimize it before publishing.

Limitations:

- Generated models may contain errors or unwanted details.
- The model may need manual editing and optimization.
- Textures, materials, or animations may not look realistic.
- Complex models may increase loading time and affect performance.
- The generated result may not exactly match the prompt.

Example: A creator generates a 3D birthday cake using AI, imports it into Lens Studio, adds balloons and a tap-to-play music interaction, and creates a birthday Lens quickly.

Q15. A fitness app wants a 3D virtual coach to show exercises next to the user in their room. Explain how world tracking and CharacterAnimation can help.

Definition: World tracking detects the phone's movement and position in a real environment, while CharacterAnimation adds movement to a 3D character.

How they help:

1. World Tracking: Tracks the phone's position and movement in the room.
2. Surface Detection: Helps place the virtual coach on a suitable surface, such as the floor.

3. 3D Coach Placement: Displays the virtual coach at the correct position and scale in the user's room.
4. CharacterAnimation: Adds exercise movements such as squats, jumping jacks and stretching to the 3D coach.
5. User Interaction: Allows users to select an exercise and start, pause or change the animation.
6. Realistic Experience: The coach appears to perform exercises inside the user's actual environment.

Example: A user opens the fitness app, places a 3D coach on the floor, and selects “Squats.” World tracking keeps the coach positioned in the room, while CharacterAnimation makes the coach demonstrate squats.

Flow: World Tracking → Coach Placement → CharacterAnimation → Exercise Demonstration

Q16. A real-estate company wants buyers to point their phone at a printed floor plan and see a 3D model of the house. Explain the tracking approach and the user experience.

Definition: Marker-based AR uses a predefined image or visual marker to recognize a real-world target and display related virtual content.

Tracking approach and user experience:

1. Prepare the Marker: Use the printed floor plan as a reference image, if it has suitable visual features.
2. Image Recognition: The phone camera detects and recognizes the floor plan.
3. Marker Tracking: Tracks the position and orientation of the recognized image.
4. Display the 3D Model: Shows the corresponding 3D house model aligned with the floor plan.
5. Interactive Exploration: Allows buyers to rotate, zoom in, or explore different parts of the model if these controls are implemented.
6. Real-Time Viewing: As the phone moves, the AR system updates the view to help buyers understand the house layout and design.

Example: A buyer points their phone at a printed floor plan. A 3D model of the house appears, allowing them to explore the living room, bedrooms, and kitchen.

Flow diagram:

Printed Floor Plan → Image Recognition → Marker Tracking → 3D House Model → Interactive Viewing

Important point: The floor plan must have enough recognizable visual features for reliable image tracking.

Q17. A gaming brand wants players to see their own Bitmoji character dancing in their room. Explain the features involved: Bitmoji, CharacterAnimation and World Tracking.

Definition: These features combine a user's personal avatar, animation and real-world tracking to create an interactive AR experience.

Features involved:

1. Bitmoji: Displays the user's personalized avatar in the Lens.
2. CharacterAnimation: Adds dancing movements to the character, such as waving, jumping or spinning.
3. World Tracking: Tracks the phone's position and movement in the room so the virtual character stays placed in the environment.
4. Character Placement: Allows the Bitmoji to appear at a suitable position and scale in the room.
5. User Interaction: Players can start or change the dance animation through available controls.

6. Immersive Experience: Combines the user's own avatar with animated movements in their real surroundings.

Example: A player opens the Lens, places their Bitmoji in the room, and taps the dance button. World tracking keeps the avatar positioned in the room, while CharacterAnimation makes it dance.

Flow: Bitmoji → World Tracking → CharacterAnimation → Dancing AR Experience

Q18. Explain the difference between Screen Text and 3D Text in a World Lens. Give an example of when to use each.

Definition: Screen Text is attached to the user's screen, while 3D Text is placed within the virtual 3D environment.

Feature	Screen Text	3D Text
Position	Fixed relative to the screen	Placed in the AR world
Movement	Stays in the same screen position	Can appear to move with the camera's viewpoint
Size	Usually remains consistent on screen	Appears larger or smaller with distance and perspective
Appearance	Looks like a normal screen label	Appears as text with a 3D position and spatial depth
Best use	Scores, instructions and notifications	World labels, virtual signs and object names

Examples:

1. Screen Text: Display “You Win!” at the top of the phone screen during an AR game.
2. 3D Text: Place the words “Welcome Home” above a virtual house model in a real-estate Lens.

Key point: Use Screen Text for information that should remain easy to read on the screen, and 3D Text for labels that should appear as part of the virtual environment.

Q19. A Lens shows a spinning 3D trophy and a “You Win!” banner on screen. Explain which camera renders each element and why.

Definition: A camera determines how visual elements are displayed in a Lens. A Perspective Camera renders 3D objects with depth, while a screen-space or orthographic setup is suitable for fixed on-screen elements.

Elements and cameras:

1. 3D Trophy — Perspective Camera: Renders the trophy as a 3D object with depth, perspective and a sense of distance.
2. Spinning Animation: The trophy's rotation can be animated to make it spin around its axis.
3. “You Win!” Banner — Screen-Space UI: Displays the banner in a fixed position on the screen.
4. Consistent Readability: The banner remains easy to read as the user moves the phone.
5. Separate Presentation: Keeping the 3D trophy and screen banner in their appropriate rendering setups prevents the banner from behaving like an object in the room.

Example: In an AR game, the trophy spins in the camera view while the “You Win!” banner stays at the top of the screen.

Remember: The trophy uses a Perspective Camera; the banner uses a screen-space UI setup. The exact camera and layer configuration depends on how the Lens is built in Lens Studio.

Q20. Explain how the Tween script works with an example, such as a logo that bounces in when the Lens opens.

Definition: A Tween script animates an object's properties smoothly over a specified time, such as position, scale, rotation or colour.

Working of Tween script:

1. Select an Object: Choose the object to animate, such as a brand logo.
2. Set Starting Values: Place the logo above its final position or give it a small scale.
3. Set Ending Values: Define the final position and size of the logo.
4. Set Duration: Specify how long the animation should take.
5. Apply Easing: Use an easing effect to make the movement appear natural, such as bouncing or slowing down.
6. Start the Tween: Trigger the animation when the Lens opens.

Example: A logo starts above the screen and moves down to its final position with a bounce effect when the Lens opens.

Flow: Starting Position → Tween Animation → Final Position

Q21. Explain, with an example, how a Behavior script can start a Tween animation when the user does something.

Definition: A Behavior script connects an event or user action to an action in a Lens, such as starting a Tween animation.

Working:

1. Create an Object: Add a virtual button, balloon or logo to the Lens.
2. Define an Event: Choose a trigger, such as a screen tap or object interaction.
3. Connect the Behavior: Configure the Behavior script to respond to the selected event.
4. Start the Tween: Set the resulting action to start the Tween animation.
5. Animate the Object: The Tween changes the object's position, scale, rotation or colour over time.
6. Show the Result: The user sees the object move or change smoothly after interacting.

Example: In a Lens game, the user taps a virtual trophy. The Behavior script detects the tap and starts a Tween that enlarges and rotates the trophy.

Flow: User Tap → Behavior Trigger → Start Tween → Animated Trophy

Q22. A tourism board wants tourists wearing Spectacles to see historical facts when they look at a monument. Would you choose AR or VR? Explain the features you would use.

Definition: Augmented Reality (AR) adds digital information to the real-world view, while Virtual Reality (VR) replaces the real-world view with a virtual environment.

Choice: AR

Features to use:

1. Real-World Viewing: AR lets tourists see the actual monument while viewing digital information.
2. Spectacles Support: Use compatible AR features available on the supported Spectacles device.
3. World Tracking: Helps position virtual labels relative to the surroundings when supported.
4. Text and Graphics: Display historical facts, construction dates and important events.

5. Spatial Placement: Position information near relevant parts of the monument, where supported by tracking capabilities.
6. User Interaction: Allow tourists to select information cards or explore additional facts using supported controls.

Example: When a tourist looks at an old fort, an AR label displays its construction year and a short historical fact beside the monument.

Why not VR? VR would replace the tourist's real view, making it less suitable for exploring an actual monument while reading information.

Q23. An ed-tech startup wants a Lens that shows a rotating solar system on a student's desk, with planet names on screen. Explain the tracking, camera and content needed.

Definition: An AR Lens combines tracking, virtual content and camera rendering to display interactive 3D objects in the real environment.

Required components:

1. World Tracking: Tracks the phone's movement and helps keep the solar system positioned on the desk.
2. Surface Detection: Detects the desk so the virtual solar system can be placed on its surface.
3. Perspective Camera: Renders the 3D Sun and planets with depth and perspective.
4. 3D Models: Add the Sun and planets, using suitable sizes, colours and textures.
5. Animation: Use rotation or Tween-based animation to make the planets orbit the Sun or rotate, as designed.
6. Planet Names: Add Screen Text for fixed on-screen labels, or 3D Text for labels positioned beside individual planets.

Example: A student points their phone at a desk. The solar system appears on the desk, the planets move around the Sun, and the planet names remain readable on the screen.

Flow: World Tracking → Desk Detection → 3D Solar System → Animation → Planet Labels

Q24. A cosmetics brand wants users to try lipstick shades and switch shades with a tap. Explain the tracking and scripting involved.

Definition: Face tracking detects and follows a user's face, allowing AR makeup effects to fit the face in real time.

Tracking and scripting:

1. Face Tracking: Detects the face and follows facial movements.
2. Facial Landmarks: Identifies the lips so lipstick can be placed accurately.
3. Lipstick Effect: Applies a virtual lipstick colour to the user's lips.
4. Screen Tap / Behavior: Detects when the user taps the screen to change the shade.
5. Script Logic: Changes the lipstick material or colour to the next shade.
6. Real-Time Preview: Updates the lipstick colour while keeping it aligned with the lips.

Example: A user tries red lipstick and taps the screen to switch to pink or nude. The lipstick colour changes without moving away from the lips.

Flow: Face Tracking → Apply Lipstick → User Tap → Change Shade

Q25. Design a festival greeting Lens (for example Diwali) that uses at least two tracking types and background music. Explain your design.

Definition: A festival greeting Lens is an AR experience that combines virtual effects, tracking and audio to celebrate a festival.

Design and features:

1. World Tracking: Places virtual diyas and decorations in the user's surroundings.
2. Face Tracking: Adds a festive glow, tilak or decorative effect to the user's face.
3. 3D Objects: Displays animated diyas, flowers and fireworks around the user.
4. Screen Text: Shows a greeting such as “Happy Diwali!” on the screen.
5. Background Music: Adds suitable festive music to create a celebratory atmosphere.
6. Animation: Uses Tween or other animation features to make the diyas glow and fireworks move.

Example: When a user opens the Lens, virtual diyas appear around the room, a festive effect appears on their face, and “Happy Diwali!” displays with background music.

Flow: World Tracking + Face Tracking → Festival Effects → Greeting Text + Music

Q26. A sports brand wants fans to raise a hand and see a virtual football appear on their palm. Explain the tracking and what happens step by step.

Definition: Hand tracking detects the user's hand and its position, allowing virtual objects to be placed on or interact with the hand.

Working steps:

1. Hand Detection: The camera detects the user's raised hand.
2. Hand Tracking: Tracks the palm's position and movement.
3. Gesture Recognition: Detects the required gesture, such as raising an open palm.
4. Football Placement: Displays a 3D football at the palm's position.
5. World Tracking: Helps maintain the AR scene as the phone moves, where supported.
6. Animation: The football can spin or glow while following the palm.

Example: A fan raises an open hand. The Lens detects the palm and places a spinning virtual football on it. When the hand moves, the football follows it.

Flow: Hand Detection → Gesture Recognition → Football Appears → Hand Movement Tracking